How Our Landing-Site Algorithm Works

Step-by-step design for the team
NASA Space Apps Challenge 2026: Lunar Landing-Site Challenge

Ash
Princess Sumaya University for Technology

October 2026

1 What the Algorithm Is Based On

We do not invent the science. The algorithm combines three things that already exist:

1. **Physics.** Sunlight and radio travel in straight lines, and the Moon has no atmosphere to bend them. So "can this spot see the Sun?" is exactly "does the straight line from here to the Sun hit any terrain?"
2. **NASA's measured data.** Terrain heights come from the LOLA laser altimeter on the Lunar Reconnaissance Orbiter. Sun and Earth positions for any hour come from JPL's ephemerides through the SPICE toolkit [3].
3. **NASA's requirements.** They define what "good" means: land within 6° of the pole, slope under 10° over 100 m, direct line of sight to Earth with no relay, and enough sunlight [1].

NASA scientists use this same "horizon method" to publish long-term illumination and Earth-visibility maps of the poles [2]. That proves the method is sound and gives us data to check against (Step 5). Figure 1 shows the full pipeline.

2 Step 0: Prepare the Terrain and Candidate Sites

- **Get the terrain.** Download the LOLA south-pole elevation map. Develop on a coarse version and move to a finer one once everything works.
- **Crop with a margin.** Candidate sites lie within 6° of the pole, but Step 1 looks far beyond each site. The terrain area must therefore be the site area plus the maximum look-out distance $D_{\max}$ on every side.
- **Fix the grid.** Use one polar-stereographic grid with a fixed pixel size. A site's ID is its (row, column).
- **Slope gate.** Compute the slope over a 100 m baseline and drop every site at 10° or steeper.
- **Candidate grid.** Start with every k-th pixel, then refine around the best areas. NASA's published average illumination map [2] can pre-select promising areas so we do not waste time on permanently dark craters.

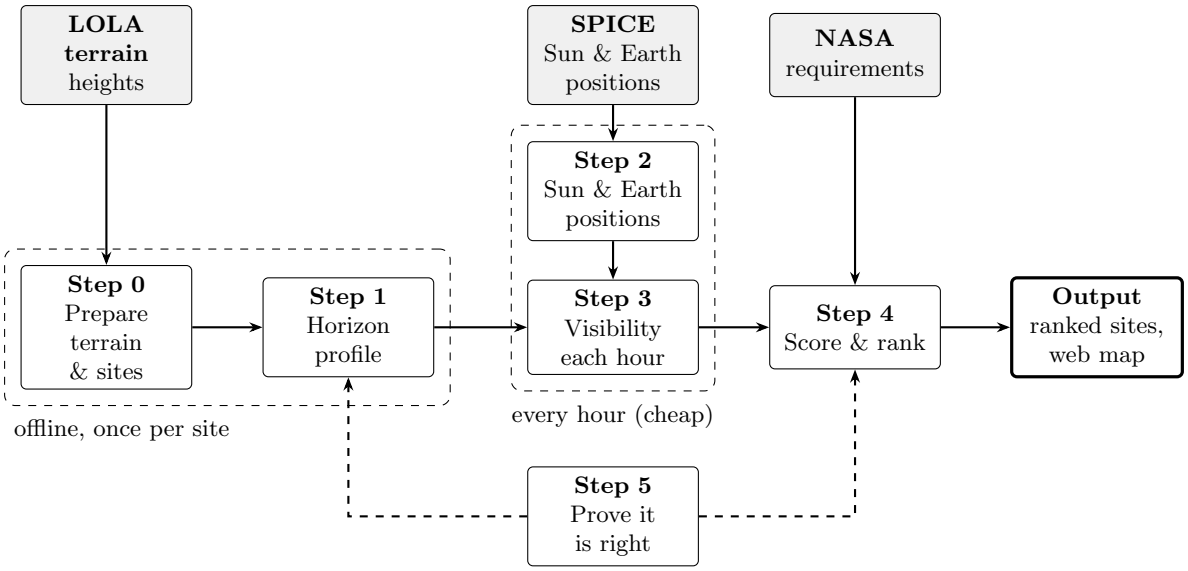


Figure 1: The pipeline. Shaded boxes are inputs from NASA. Steps 0–1 run once, offline; Steps 2–3 are a fast lookup for any dates the user picks. Step 5 checks the results of Steps 1 and 4.

Direction convention (critical). At the south pole every direction is "north", so compass bearings do not work. All steps measure azimuth θ from the map's $+x$ axis, counterclockwise, 0–359°. Steps 1 and 2 must use exactly this convention.

3 Step 1: Horizon Profile (the Hardest Step)

Stand at a site and look in direction θ. Walk outward over the terrain. At each distance d, compute the angle up to the terrain. The steepest angle found is the horizon $H(\theta)$ in that direction (Figure 2). Repeat for all 360 directions to get the site's full skyline.

$$H(\theta) = \max_{d \le D_{\max}} \arctan\left(\frac{z(d) - (h_0 + m) - d^2/2R}{d}\right), \qquad R = 1737.4 \text{ km} \tag{1}$$

Here $z(d)$ is the terrain height at distance d (interpolated between pixels), h_0 is the site height, m is the mast height, and $d^2/2R$ is how far the Moon's curved surface drops below a flat line. The curvature term matters: at 20 km it is about 115 m, which shifts the angle by about 0.3°. That is significant when the Sun is only about 1.5° up.

Algorithm 1 Horizon profile for one site

Require: terrain Z, site (x_0, y_0), height h_0, mast m, look-out distance $D_{\max}$

for $\theta = 0, 1, \ldots, 359$ **do**
  $H[\theta] \leftarrow -90°$; $d \leftarrow d_{\min}$
  while $d \le D_{\max}$ **do**
    $(x, y) \leftarrow (x_0 + d\cos\theta,\ y_0 + d\sin\theta)$
    $z \leftarrow \textsc{Interpolate}(Z, x, y)$
    $\alpha \leftarrow \arctan\big((z - h_0 - m - d^2/2R)\,/\,d\big)$
    $H[\theta] \leftarrow \max(H[\theta], \alpha)$
    $d \leftarrow d + \textsc{Step}(d)$ ▷ small near the site, larger far away
  end while
end for
return H

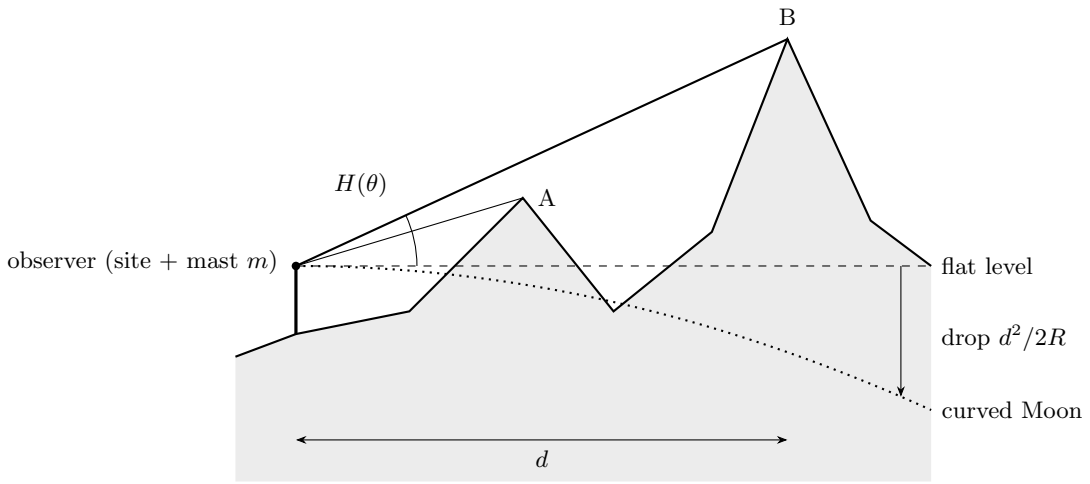


Figure 2: Step 1 in one direction. Keep the steepest angle to the terrain. Peak A is nearer; peak B is farther but higher, and it sets the horizon because its angle is larger. Heights and curvature are exaggerated.

Why compute the horizon first. Checking the Sun directly would need one walk over the terrain for every hour: 8,760 walks per site for a year. The horizon needs 360 walks once, and after that every hour is a lookup. That is about 24 times less work, and trying new dates costs almost nothing.

What makes it hard. The total cost is (sites) $\times$ 360 $\times$ (steps per walk). With 10,000 sites and 2,000 steps, that is about 7 billion terrain samples. It needs vectorised NumPy or Numba, parallel runs across sites, and a step size that grows with distance. Run it at a few mast heights (for example 0, 2, 5 and 10 m) so Step 4 can recommend a mast.

4 Step 2: Sun and Earth Positions

For every site and every hour, use SPICE (through the SpiceyPy library [4]) to get the direction of the Sun and Earth as seen from that site: azimuth $\mathrm{az}(t)$ in our map convention, and elevation $\mathrm{el}(t)$ above the local flat level. The kernels needed are the leap-second file, a planetary ephemeris (for example DE440), and the Moon's orientation kernels.

The Sun and Earth move very differently (Figure 3):

- **Sun:** goes all the way around the horizon once a month, staying within about $\pm 1.5^\circ$ of level. So sunlight depends on the skyline in *every* direction.
- **Earth:** stays in one narrow direction and bobs up and down by about $\pm 6.7^\circ$ over about 27 days. So Earth contact depends on the skyline in *one* direction only.

These ranges double as sanity checks. If SPICE output falls outside them, something is wrong.

5 Step 3: Visibility Each Hour

For each site and hour t, compare the sky positions with the horizon. H is interpolated between whole degrees. The Sun and Earth are discs, not points (radius $r_\odot \approx 0.27^\circ$, $r_\oplus \approx 0.95^\circ$ as seen

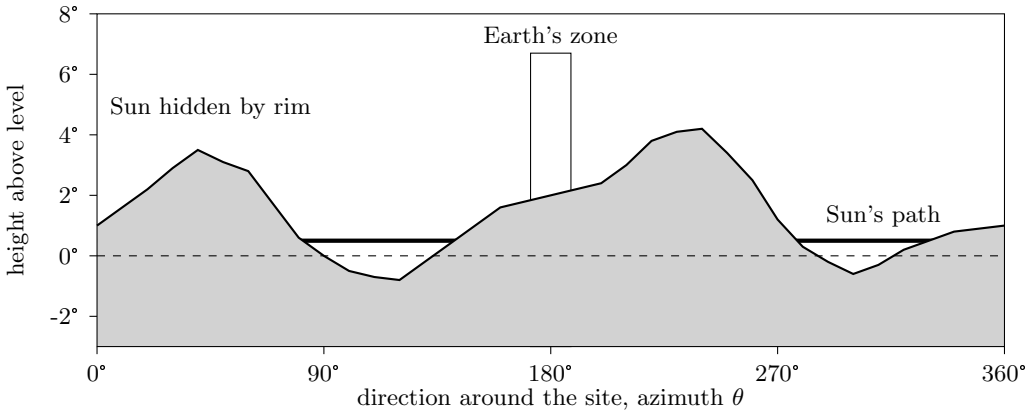


Figure 3: Steps 2 and 3 in one picture (illustrative terrain). Gray is the horizon profile from Step 1. The Sun moves along the flat line and is visible only where the line is above the gray. Earth stays inside the narrow band and moves up and down; it is visible when it rises above the gray at 180°.

from the Moon):

$$s(t) = \mathrm{clamp}\left(\frac{\mathrm{el}_\odot(t) - H(\mathrm{az}_\odot(t)) + r_\odot}{2\,r_\odot},\ 0,\ 1\right) \qquad \text{fraction of the Sun visible (power)} \qquad (2)$$

$$e(t) = \left[\,\mathrm{el}_\oplus(t) + r_\oplus > H(\mathrm{az}_\oplus(t))\,\right] \qquad \text{any part of Earth visible (comms)} \qquad (3)$$

$$b(t) = \left[\,s(t) > 0.5\,\right] \wedge e(t) \qquad \text{power and comms together} \qquad (4)$$

The "any part of Earth" rule matches the one NASA uses for its Earth-visibility maps [2]. Figure 4 shows what Step 3 produces for one site.

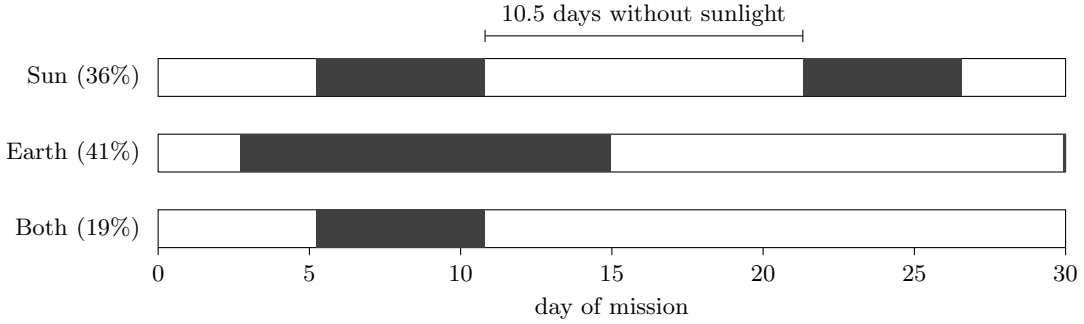


Figure 4: Step 3 output for one site over 30 days (illustrative). Black means visible. Sun and Earth are each visible about 40% of the time, but both together only 19%, and there is a 10.5-day stretch without sunlight that the battery must survive. A monthly average hides all of this.

6 Step 4: Score and Rank

Gates. A site must be within 6° of the pole and have a slope under 10°. Sites that fail are removed, not scored.

Metrics over the mission window T. f_{sun}, f_{earth} and f_{both} are the fractions of hours with $s > 0.5$, e and b true. g is the longest stretch without sunlight.

$$S = w_1 f_{\mathrm{sun}} + w_2 f_{\mathrm{earth}} + w_3 f_{\mathrm{both}} - w_4\, g/T \qquad (5)$$

The weights are set by the user in the web app.

Battery survival. A simple energy budget with panel power P_{panel}, load P_{load} and time step Δt:

$$E_{t+1} = \min\bigl(E_{\max},\ E_t + (P_{\text{panel}}\, s(t) - P_{\text{load}})\,\Delta t\bigr) \tag{6}$$

The site survives if $E_t \geq E_{\min}$ at every hour. Sites that do not survive are ranked last.

Landing accuracy. A lander cannot hit an exact pixel; NASA's reference is within 100 m of the target [1]. So rank each site by the worst score within 100 m of it:

$$S_{\text{robust}}(p) = \min_{|q-p|\leq 100\ \text{m}} S(q) \tag{7}$$

Mast height. Using the horizons from Step 1 at several mast heights, report the lowest mast that meets the targets.

When to land. Repeat the scoring for every possible start date. The result is a calendar showing the best arrival dates for each site.

7 Step 5: Prove It Is Right

1. **Smooth-Moon test.** On a perfectly smooth sphere, an observer at height h must see the same horizon in every direction: $H = -\arccos\bigl(R/(R+h)\bigr)$, about $-0.087°$ for $h = 2$ m. This also checks that $D_{\max}$ reaches past the tangent distance $\sqrt{2Rh} \approx 2.6$ km.
2. **Single-cone test.** One cone-shaped peak of height P at distance L must give $H = \arctan\bigl((P - h_0 - L^2/2R)/L\bigr)$ in its direction.
3. **Look-out distance.** Run with $D_{\max} = 50$, 100 and 150 km. Increase it until the horizon stops changing by more than about 0.02°.
4. **Compare with NASA's maps.** NASA's illumination maps [2] are long-term averages, so compare them with our results averaged over at least a year, not one month. The same sunny ridges should stand out, such as the rim of Shackleton crater.
5. **Check the sky.** The Sun stays within about ±1.5° of level and Earth within about ±6.7° (Step 2).

8 Files Passed Between Steps

Everyone builds against fake files in these formats from day one, so nobody waits for anyone. Angles are in degrees, times are hourly UTC, and azimuth follows the convention in Step 0.

File	Contents	Shape	Made by	Used by
`dem.npy`	Terrain heights (m) on the map grid	rows × cols	Step 0	Step 1
`sites.csv`	id, x, y, lat, lon, height, slope	one row per site	Step 0	Steps 1, 4
`horizon.npy`	Horizon profile per site and mast height	sites × masts × 360	Step 1	Step 3
`sky.npz`	Sun and Earth az/el each hour	sites × hours	Step 2	Step 3
`visibility.npz`	$s(t)$, $e(t)$, $b(t)$	sites × hours	Step 3	Step 4
`results.json`	Scores, metrics, timelines of top sites	per site	Step 4	Web app

Table 1: Data contracts between steps.

9 Who Owns Which Step

Step	Role	Member
Step 0: terrain and sites	Terrain and sites	_____
Step 1: horizon profile	Horizon engine	_____
Step 2: Sun and Earth positions	Sky (SPICE)	_____
Steps 3–4: visibility and scoring	Scoring and mission model (integrator)	_____
Display	Web app	_____
Step 5 and submission	Validation, story and submission	_____

Table 2: Step ownership. The horizon engine owner also writes the smooth-Moon and cone tests.

10 What Makes Our Tool Different

NASA already publishes average illumination and Earth-visibility maps [2], and judges may know them. Averages cannot answer a mission's real questions, and Figure 4 shows why. Our tool answers them for a specific mission:

- When will this site have power and Earth contact *at the same time*?
- Will a lander arriving on a given date survive its whole mission?
- How tall must the mast be at this site?
- Is the site still good if we land 100 m off target?
- Which arrival date is best?

References

[1] NASA, *Artemis Overview, 2025 International Space Apps Challenge*, NTRS 20250009323. https://ntrs.nasa.gov/api/citations/20250009323/downloads/ArtemisOverview2025IntSPAPPSChallenge.pdf

[2] NASA Goddard Planetary Geodesy Data Archive, *Lunar Polar Illumination*. https://pgda.gsfc.nasa.gov/products/69

[3] NASA NAIF, *The SPICE Toolkit*. https://naif.jpl.nasa.gov/naif/

[4] SpiceyPy, Python wrapper for SPICE. https://github.com/AndrewAnnex/SpiceyPy